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Business Cycles from Misperceived Trends

D. Heymann and P. Sanguinetti ¹

1. Introduction and Overview

Much of the current macroeconomic literature uses an explicit intertemporal framework. However, little attention is usually given to the matter of how, and if, the plans that different individuals draw into the future are reconciled with one another. The assumption that, whatever are the features of the environment, agents form their expectations as if they knew the processes generating future outcomes makes it difficult to identify even the possibility of problems in achieving intertemporal coordination (cf. Leijonhufvud (1981, 1993)): the path of the decision variables can be disturbed by exogenous random events, but there is no room for decision errors due to incorrect forecasts of market behavior. The coordination of future actions would thus result automatically, not through the operation of a market process, but from the assumed properties of individual expectations. Models of intertemporally coordinated economies can in fact generate non-trivial dynamics, such as fluctuations in consumption, aggregate demand and the current account in open economies subject to certain impulses. However, these results are often obtained from exercises where a system undergoes an arbitrary shift in parameters (e.g. a change in the policy regime) when, in fact, the rational expectations assumption logically rules out the possibility of such events (Sargent (1993)). From a practical point of view, it seems difficult to deny a priori the possibility that in some occasions, agents find themselves faced with new scenarios, where they have to revise their beliefs and therefore reformulate their previous plans.

The view that misperceptions about the future real consequences of current decisions can give rise to business fluctuations has a long tradition in Macroeconomics. Booms and contractions have often been interpreted as originating from swings of opinions that cause, for example, an over-expansion in aggregate demand eventually followed by a credit crunch. Theories which allow for intertemporal coordination failures can take quite different specific forms. They have in common the position that agents decide on the basis of less than full knowledge of how the environment evolves, and that the resulting difficulties in forming accurate expectations can sometimes have noticeable macroeconomic consequences. Such difficulties are likely to

¹ CEPAL, Instituto Di Tella and Univ. of Buenos Aires, and Univ. Torcuato Di Tella, respectively. Parts of the paper were written while D. Heymann was visiting at the University of Trento. Special thanks are due to P. Azcue for the advice and help that he gave for performing the simulations which are reported in section 2 of the paper. The comments received from F. Guerrero, A. Leijonhufvud and J. P. Nicolini are gratefully acknowledged. The authors are responsible for the errors and the opinions contained in the paper.

become especially relevant when individuals cannot rely on the validity of behavior patterns that they have observed in the past. Accordingly, the analysis of problems of intertemporal coordination commonly takes as a point of departure a more or less sudden shift that disturbs the growth path that the economy has been following. It is the inability of at least some agents to anticipate the effects of this change which generates a business cycle episode.

In this class of theories, therefore, there is a close connection between imperfect knowledge about growth patterns and the emergence of cyclical ups and downs. Of course, in many instances it can safely be assumed that, since the growth process is taking place more or less smoothly, individuals can extrapolate past performance in order to predict their future opportunities with some accuracy. However, there are also important cases where the assumption seems unwarranted. Events such as policy reforms, external shocks or technical changes can drastically alter the behavior of a given economy, but often there is scarce theoretical or empirical evidence to anticipate the particular features of the new growth path. In such circumstances, although agents may try their best in forming expectations, and base their plans on the realization (or the belief) that a break in the trend has taken place, this does not imply that one should attribute to them an ability to make precise forecasts that neither theory nor past experience provide ².

Although these issues are of a general nature, they may be relevant for the analysis of specific episodes. In the Latin American experience of the last decades, for example, there were a number of instances in which economies showed large swings in aggregate spending, relative prices and the trade balance, with an initial phase of rapid growth in domestic demand followed by a sharp contraction; in some cases, these cycles started with wide-ranging policies of stabilization and reforms. It is possible to view post-reform fluctuations in different ways. Indeed, initial increases in demand, coupled with real appreciations and trade deficits, can be rationalized qualitatively as an equilibrium response to good news (cf. Feliz (1994), Maia and Ortiz (1995), Uribe (1995) and Roldos (1995)). Another related line of argument stresses the incomplete credibility of policies: if, say, individuals perceive that an exchange rate-based disinflation will be short-lived, they may plan to change the time profile of consumption, with an increase during the period in which the inflation tax is low and a (pre-programmed) decline when the inconsistent policies are abandoned and inflation resumes (cf. Calvo (1986), Kiguel and Leviatan (1992), Mendoza and Uribe (1996), Talvi (1995)). Such models can generate cycles, but they do not account easily for the gradual rise in spending that is often observed (Rebelo and Vegh (1995)). In addition, since the argument is based on the presumption that agents correctly foresee the future path of both economic variables and of government behavior, it would not hold when policies turn out in fact to be durable.

In the two modes of analysis mentioned before, the cycles which may arise are

²Economists routinely meet to discuss the lessons to be drawn from one set of episodes or another. If such exercises are found useful, it looks difficult to argue without contradiction that, whatever the circumstances, agents have nothing to learn.

assumed to be anticipated correctly by agents: the recession, if and when it comes, is part of a path that individuals have knowingly chosen. These models allow for sudden changes in certain variables (e.g. an abrupt fall in foreign reserves preceding a devaluation, as in Krugman (1979)), but not for crises in the common meaning of the term, that is, for events that disturb the plans of agents. Another body of literature has concentrated on the possibility that crises may be modelled as sunspot phenomena, i.e. as changes in behavior induced by the coordination of individual expectations on the basis of some random event which is not part of the system's fundamentals (cf. Calvo (1996), Kehoe (1996), Velasco (1996)). Here, the downturn is unanticipated, but it results from a well-coordinated (albeit possibly suboptimal) behavior. Although the sunspot models can rationalize the sudden drop in the demand for domestic financial assets that is typically observed in crises, the question naturally arises about how agents may focus their expectations unanimously on a particular set of events. Moreover, it is unclear whether one should assume that the coordination of expectations arises spontaneously once the event that triggers the crisis has taken place, or that the agents have been somehow predicating their choices on the expectation that the crisis would occur given the event, and have included the correspondent contingencies (with their objective probabilities) in their plans. In this second case, the crisis is simply due to the realization of a shock. In the first scenario, the sunspot argument implies the existence of misperceptions, since agents have not considered in the past decisions the likelihood that a crisis may erupt.

Our aim in this paper is to consider some classes of business cycles in economies undergoing rapid changes in their real configuration, without concentrating specifically on the moments of crisis. In this regard, our analysis may be seen as complementary with some of the literature that studies disruptions in financial markets. We start from the view that those disruptions are more likely to appear in economies in which agents have seen their expectations disappointed: phenomena such as runs do not come out of the blue. The hypothesis that we make is that some fluctuations can derive from an over-reaction of agents to fundamental news that raise the economy's real output. When individuals overpredict future incomes, consumption will rise above its sustainable value. Investment will do likewise if firms entertain exaggerated expectations of the return to capital. In an open economy, this behavior will be reflected in an excessive current account deficit. Thus may arise a cycle: eventually, the mistakes are revealed as being such, and spending adjusts (cf. Heymann (1983, 1994), McKinnon and Pill (1994)).

We explore this possibility through several analytical exercises. The framework that we utilize has many points in common with that of the real business cycle literature for open economies in intertemporal equilibrium. The fluctuations that we deal with are of a real nature, in the sense that they originate from a shift in real opportunities, and the behavior of the system is studied without imposing restrictions on the adjustment of prices. However, we differ from the equilibrium analysis by treating future income and the profitability of investment (and, therefore, wealth) as variables that agents have to forecast through some inference procedure or learning algorithm, which will

not automatically make them knowledgeable of the values (or, for that matter, the probability distributions) that the system will in fact deliver.

We want to represent forward-looking agents who fall short of being all-knowing. The individuals that we model recognize when there has been a change in fundamentals (even if they do not know the economy's true model). It is precisely this property of agents's forecasts, and the fact that non-equilibrium behavior can be confused for a movement along a coordinated path which sustain for some time the misperceptions that generate the spending cycle.

In section 2, we develop an exercise where there is initially a shock that increases (permanently) the productivity of capital in the traded-goods sector of an open economy. We assume that there are externalities in production in this sector, so that the rate of return for each agent (and therefore, his/her desired investment and perceived wealth) depends on aggregate output. Individual plans are thus predicated on expectations about the economy's performance. The model is built in such a way that, under perfect foresight, aggregate consumption would jump immediately to a new constant path (due to an upward shift in wealth perceptions), while the capital stock and aggregate output gradually converge to higher steady-state levels. Agents are supposed to be aware that there has been a positive shock on output, and that actual aggregate production will relax towards a new value. However, they do not know precisely the numerical value of the aggregate steady-state and the speed of convergence to it.

These parameters of the new growth path are more or less intricate functions of the aggregate productivity shock, of the features of the production functions, and of the aggregate consequences of individual expectations. We assume that agents do not engage in the complicated task of trying to estimate a structural model (and we do not wish to make the a priori hypothesis that the parameters of that model are somehow public information). Rather, we suppose that, starting from some initial guess, agents revise their beliefs about the parameters of the growth path by using an error-correction procedure based on the difference between the anticipated and the actually observed levels of output: if current production is higher than expected, individuals revise upwards their estimates of steady-state aggregate production. Such a learning scheme can be considered as a simple member of the class of time-series learning algorithms (cf. Sargent (1993)).

Clearly, the behavior of the system depends on the initial beliefs of agents, that is, on their starting judgments about the likely effect of the shock. But, in addition, the process of learning about the two-parameter growth path of traded-goods output can make the estimates of these two numbers diverge for some time from their true values, rather than induce them to move monotonically towards them. In the exercise presented in section 2 we focus on the case where agents initially overestimate the steady state level of aggregate output, and they underestimate the speed of convergence. This would correspond to a situation where individuals perceive that the economy has received very good long-run news, but, at the same time, they believe that the effects on aggregate output will materialize slowly. In those circumstances,

the first realizations of output may pleasantly surprise individuals, and cause them to raise further their estimates of steady-state production, on which wealth perceptions depend. The exercise shows that aggregate consumption can then be unsustainably high, and nevertheless keep increasing for a certain number of periods, before it starts declining.

It is reasonably straightforward to extend the model to a two- good setting, allowing for the existence of non-traded goods. In the two-sector case, individuals, both as producers of non-traded goods and as consumers, must make anticipations about the real exchange rate, and therefore, they have to learn about this variable. For the purpose of representing relative price forecasts we assume that, even though agents do not know the full model of the economy in its details, they understand the link between the real exchange rate and the supply/demand of non-traded goods. That is, each agent in our model expects the real exchange rate to move (perhaps gradually) to a conjectured sustainable value, which he/she re-calculates every period, and which is consistent with the forecast that the agent is simultaneously making of the future output of traded goods (and, therefore, of aggregate wealth). In this regard, individuals are guided by their perceptions of the economy's fundamentals. However, since these are unknown, the system can generate non-coordinated outcomes. In particular, exaggerated expectations about the steady-state aggregate output of traded goods lead to excessive real appreciations, which can not only persist for some periods but also grow over a certain time interval: it is possible that the outcomes observed soon after the productivity shock make agents increase their already too-high estimates of the sustainable relative price of non-traded goods.

No claim will be made about the uniqueness of the learning schemes which are postulated in the exercise (and, in fact, one may expect that there will be typically much heterogeneity in the way reasonable individuals interpret observed economic behavior). However, the results would indicate that, although the economy eventually converges to a steady-state where outcomes and perceptions are matched with one another, learning need not proceed monotonically. Excessive optimism can be temporarily self-reinforcing, so that at first wealth perceptions diverge increasingly from what would be the correct values. Therefore, ignorance about the trend of real income may generate transitional cycles in aggregate spending and relative prices. In an open economy, such cycles would have lasting consequences, through their effect on the level of the foreign debt.

In the model just sketched, there are two superimposed dynamics: that generated by the consumption and investment decisions of agents and that induced by the change in their beliefs. Learning proceeds on the basis of initial assumptions about the features of the paths of relevant variables and their observed evolution over time. In section 3 we take another approach to the analysis of the possible ways in which individuals make inferences about the aggregate performance of the economy. The question we ask is whether agents who process aggregate information through a (simplified) structural model will immediately correct misperceptions or, by contrast, these can be self-sustaining. In the example we present, the answer turns out to depend on

the opinions individuals entertain about the beliefs of others. More specifically, it is conceivable that, if agents consider that market outcomes result from decisions based on accurate forecasts on the part of the average trader, they would interpret, say, an observed increase in aggregate consumption as an indication that future total income will in fact be higher. Then, if the future income of the agent somehow depends on the aggregate, he would be led to raise his own current consumption, even though every individual is just reacting likewise to what he perceives as a reliable market signal, when actually there has not been a corresponding shift in fundamentals. Although the result depends on some particular hypotheses, the exercise suggests that in some cases, a strong belief on the part of agents that aggregate behavior is predicated on correct expectations can induce them in fact to make mistaken choices, which will be revealed as such in the future.

The arguments discussed in section 3 have analogies with those of the expectations of expectations literature (cf. Frydman and Phelps (1983)), and with the more recent studies on informational cascades (cf. Banerjee (1992), Bikchandani et al. (1992), Hirschleifer (1995), Lux (1995)). They have also points in common with the analysis of over-borrowing episodes presented by McKinnon and Pill (1994). In that paper, however, banks can form exact predictions (while the general public misreads the economy's signals by mistakenly attributing an increase in banks's willingness to lend to the expectations of intermediaries instead of realizing that it derives from the incentives produced by guarantees on deposits) whereas, in our case, there is no necessarily an asymmetry in the accuracy of the expectations of different classes of agents: in the example just presented, all individuals can maintain biased beliefs. Irrespective of their detailed specification, our analysis suggests the possibility that in systems where, say, over-optimistic expectations somehow develop, there may be departures (not immediately self-correcting) from the fundamental path. These, however, would be different from the traditional rational bubbles, since they would derive from an incorrect interpretation of the actions and beliefs of other agents, instead of being the result of a particular set of coordinated expectations. In the scenarios we explore here, the existence of a bubble leads to a future revision of plans.

2. Learning about the steady-state

We are interested in exploring the evolution of income and wealth perceptions in a system where the individuals must make conjectures about the future performance of the economy in the aggregate. When posed in general terms, the analytical problem seems quite complicated. More or less drastic simplifications appear to be called for. Although we are concerned with the interaction between growth and cycles, the model that we present here is built so that the system converges to a steady-state with constant output. This feature does not look too restrictive: we can still study the dynamics during the traverse between two steady states, and the model seems apt to be extended in order to allow for endogenous growth. The economy is described through a representative agent construction. The use of this modeling device is not

meant to imply that it is common knowledge that all individuals behave identically. Rather, the representative agent happens to be the average individual of the system, with no presumption that he (or anyone else) acts as if the behavior of the economy was a scaled-up version of his own (cf. Leijonhufvud (1974)).

The model does not contain any element of exogenous uncertainty³. Therefore, along the equilibrium path, agents behave with perfect foresight. However, the economy need not stay along that path, since individuals lack prior knowledge of all the relevant parameters. In order to solve the model while learning is under way, some assumptions have to be made regarding how agents form their expectations, and the degree of confidence they have on those. We will endow agents with a learning procedure but, for the sake of simplifying the analysis, we will represent their decisions as producers and consumers as if, at each point in time, they had no doubts about the accuracy of their expectations. That is, individual plans will be derived from deterministic, and not probabilistic optimization problems⁴.

2.1. A simple one-good setup

We consider as a first step a one-good open economy, which faces a perfectly elastic supply/demand of foreign credit at an interest rate that equals the rate of time preference of the representative individual. The model is specified in a very standard fashion. The economy is populated by infinitely-lived individuals who produce and consume the good, which is perishable. Intertemporal budget constraints are automatically binding. The preferences of the representative agent are assumed to be time-separable,

$$U_{it} = \sum_{j=1}^{\infty} \beta^{j-t} U_i({}_t c_{ij}) \quad (1)$$

We have supposed that agents decide as if they faced no uncertainty. The specification of preferences is such that the individual will choose a consumption path which gives a constant expected consumption. Therefore, ${}_t c_{ij}$, the planned consumption in period j chosen by individual i in period t , will be equal to the return on perceived wealth:

$${}_t c_{ij} = {}_t W_{it}^i (1 - \beta) = \frac{r}{1+r} {}_t W_{it}^i, \quad j = t, t+1, \dots \quad (2)$$

³However, there is an initial impulse that disturbs the previous steady state, and triggers are transitional dynamics that we want to analyze. The shock is assumed to be fully unexpected. The assumption does not seem to imply a logical contradiction, since we deal with agents who do not necessarily have the ability to predict future outcomes.

⁴This simplification implies that the analysis leaves aside the effects of risk aversion and flexibility preference. In the initial stages of an economic transition, it is likely that agents will recognize that their forecasts are uncertain, and will therefore choose to delay commitments until they have more reliable expectations of how the system will evolve. Such behavior would moderate the impact of optimistic expectations on actual spending. However, this can influence individual perceptions, if agents believe that the aggregate outcomes that they observe result from a "prudent" behavior of others. The point is discussed from another perspective in section 3.

As it is well known, consumption according to (2) implies that the individual plans to maintain a constant level of wealth.

Each individual produces the good using capital and a fixed input. We assume that there are externalities in production, so that the output of a given individual varies positively with the aggregate level of output⁵. Production also depends on a shift variable, which we will interpret as depending from features of the economic environment determined, for example, by economic policies. The production function is then specified as (with y_{it} being the output of individual i in period t , z_{it} the shift parameter, K_{it-1} the beginning-of-the-period capital stock held by individual i in period t , and y_t aggregate output):

$$y_{it} = z_{it} K_{it-1}^{\alpha} y_t^{\eta} \quad (3)$$

As it is common in the literature, we suppose that changes in the capital stock have associated adjustment costs. For simplicity, these are taken to be a symmetric (quadratic) function of the gradient of the capital stock during the period. The productive decisions of the individual are governed by the objective of maximizing present value of net output. This results in the following problem:

$$\max \sum_{j=t}^{\infty} \beta^{j-t} \left[{}_t y_{ij} - ({}_t K_{ij} - {}_t K_{ij-1}) - \frac{\phi ({}_t K_{ij} - {}_t K_{ij-1})^2}{2 {}_t K_{ij-1}} \right] = {}_t W_{it}^{iy} \quad (4)$$

Here ${}_t y_{ij}$ denotes the level of individual output in period j planned at t by agent i (and similarly for ${}_t K_{ij}$), ${}_t W_{it}^{iy}$ indicates the present value, perceived by individual i , of his production plan (net of investment and adjustment costs) set at t .

Due to the externality, the path for the capital stock, the value of perceived wealth and consumption will depend on the expectations that the individual forms about aggregate output. Given the accumulation and production plan derived from the programming problem (4), and given those expectations, the individual estimates the present value of the planned flow of net (of investment and adjustment costs) output. Wealth equals that present value of net output, less the agent's financial liabilities:

$${}_t W_{it}^i = {}_t W_{it}^{iy} - (1+r)b_{it-1} \quad (5)$$

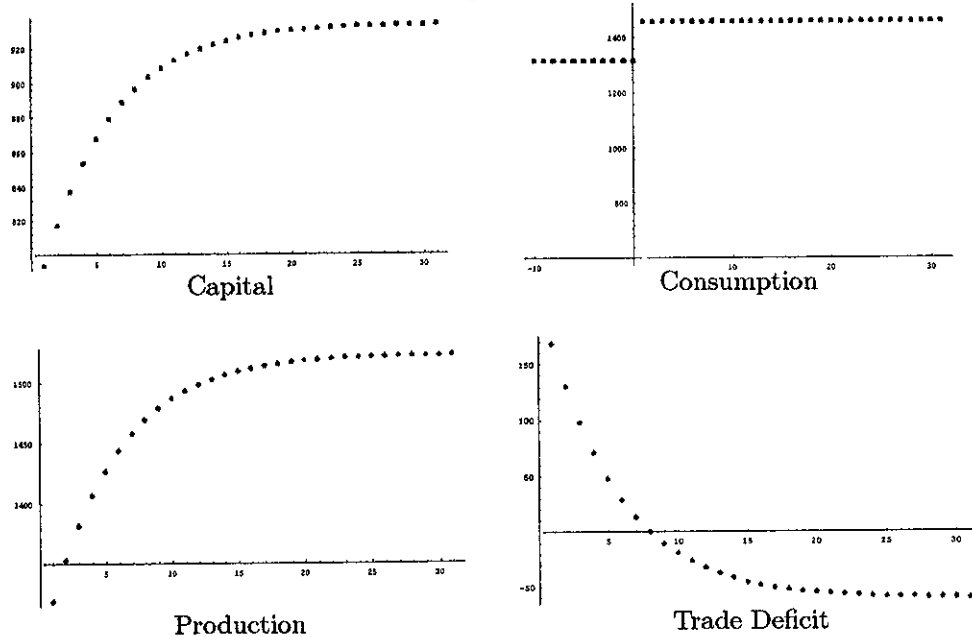
We assume that initially (at $t = 0$), those liabilities are zero; thereafter, they evolve according to the difference between spending and output:

$$b_{it} = b_{it-1}(1+r) + c_{it} + (K_{it} - K_{it-1}) + \frac{\phi (K_{it} - K_{it-1})^2}{2 K_{it-1}} - y_{it} \quad (6)$$

⁵We use this formulation as a straightforward way to represent the (commonsense) assumption that the opportunities of a given agent depend on the aggregate performance of the economy. The influence of total output on individual production may derive from technology or from other effects such as "thick market" externalities (cf. Howitt and McAfee (1992)). In the two-good model that we present below, the real income of an individual depends on the economy-wide production of traded-goods (through the demand for non-tradeables), even in the absence of externalities.

The system defined by expressions (1)-(6) determines the production, consumption and financing plans of the individual for a given set of expectations. A perfect foresight solution can be obtained by making anticipated future output equal to the value that the model delivers for each period. For the purpose of this discussion, we have assumed that the externality in production is not too strong (i.e. $(\alpha + \eta) < 1$), so that the perfect foresight paths for output and the capital stock converge to a steady-state⁶. **Figure 1** show the simulated paths of the main variables after an upward (permanent) productivity shock which raises steady-state output by 15%.

Figure 1



These solutions have standard qualitative features. The adjustment costs in investment delay the increase in the capital stock, and cause it to be gradual. However, starting from a previous steady-state, wealth perceptions are immediately revised after the arrival of the news about the productivity shift. Consumption thus moves at once to the new steady-state level. During the first periods of the transition, investment is also high, while production is yet far from the new steady-state. The system then shows trade and current account deficits, as foreign credit is used to finance the higher level of spending. Along the transition path, aggregate expenditure falls (due to the decrease in investment), while output increases. Eventually, the trade balance becomes positive. As the economy approaches the steady-state, the trade surplus

⁶We have used this property to solve the optimization problem (1)-(6). The first order conditions for this problem define a non-linear difference equation. We obtained approximate solutions numerically as the roots of a system of equations, assuming that the steady state is reached after 40 steps.

becomes close to the value of the interest services on the accumulated debt. Wealth remains constant after being re-evaluated when the shock is observed; as should be the case on a perfect foresight path, this validates the expectations that sustained the initial jump in consumption. For the sake of the discussion of learning procedures (see below), it may be worth noting that the sequence of aggregate output generated by the model can be represented, with a good approximation, by a first order autoregressive process of the following form,

$$y_t = \lambda y_{t-1} + (1 - \lambda)\bar{y} = \lambda y_{t-1} + y^* \quad (7)$$

According to expression (7), aggregate output in a given period is a convex combination of its value in the previous period and of the new steady-state output $\bar{y}$. The parameter λ is a measure of the speed of the transition: if it is small, the autoregressive term is also small, and consequently output relaxes fast to its steady state.

2.2. An extension: traded and non-traded goods

One of the main simplifications made in the previous analysis was the assumption that the economy produces a single good. The dynamics of relative prices were therefore left out of the picture. However, these dynamics can be introduced through a simple extension. We now assume that the representative agent produces a traded good (T) and a non-traded good (N). The output of traded goods is specified as before. That means, in particular, that good N does not enter as an input into the production of good T . This makes the production and capital accumulation of the traded goods sector independent from the movements in relative prices. The specification does not of course describe a general case, but it still allows us to study the consequences of price expectations on consumption decisions, which is an effect we want to concentrate on. The non-traded good, we assume, is only used for consumption, and it is perishable. Intertemporal preferences are of the same type as before and we assume that preferences between the two goods are described by a Cobb-Douglas specification, so that in a given period the agent allocates the value of consumption in fixed proportions to each good. In what follows, we will denote by h the proportion of total consumption corresponding to expenditures in non/traded goods. Good N is produced with traded inputs and a fixed factor; there is no accumulation of capital in the sector producing non-traded goods ⁷. The production function of good N is given by:

$$y_{Nij} = A x_{ij}^{\epsilon} \quad (8)$$

⁷We are in fact modeling a case of unbalanced growth during the transition, since only the traded-goods sector receives a productivity shock and accumulates capital. It may be noted that, if investment in sector N was also considered, the inference problem of agents would become quite complicated. A producer of good N would have to forecast the aggregate investment in that sector, and evaluate the effects of the increase in capacity on the price sequence of the good (cf. Heymann (1983), (1984)).

where y_{Nij} and x_{ij} are, respectively, the production of N by individual i in period j , and the use of (traded) inputs in that period by that individual.

The input demand and the output supply functions corresponding to (8) are:

$${}_t x_{Nij} = ({}_t p_{Nj}^i A \delta)^{\frac{1}{1-\delta}} \quad (9)$$

$${}_t y_{Nij} = A^{\frac{\delta}{1-\delta}} \zeta^{\frac{\delta}{1-\delta}} {}_t p_{Nj}^{i \frac{1-\delta}{1-\delta}} \quad (10)$$

In the previous expression ${}_t p_{Nj}^i$ denotes the price of non-traded goods in period j , as perceived by individual i in period t and ${}_t y_{Nij}$ the planned (in t) supply of goods N in j . The markets are assumed to operate in the following way. After the agent has observed aggregate output and the relative price in period $t-1$, he formulates a forecast of the future path of those variables (i.e. expectations for the variables at $t, t+1, t+2, \dots$). Using those expectations, he decides (as part of an intertemporal plan) the levels in period t of investment in sector T , the value of consumption of each good, the demand for inputs in the production of N , and the resulting output of N . These magnitudes are realized in period t . The market for N clears: the price of non-traded goods is established so that the value of output equals that of consumption (see flow diagram 1).

In order to choose his level of consumption, the individual must evaluate his wealth; this depends on the anticipated price of non-tradables. If traded goods are used as numeraire, the wealth of the agent as perceived at the beginning of period t (i.e. after observing variables in $t-1$ but not in t) would be given by:

$${}_t W_{it}^i = \sum_{j=t}^{\infty} \beta^{j-t} \left[{}_t y_{ij} - ({}_t K_{ij} - {}_{t-1} K_{ij-1}) - \frac{\phi ({}_t K_{ij} - {}_{t-1} K_{ij-1})^2}{2 {}_t K_{ij-1}} \right] + \sum_{j=t}^{\infty} \beta^{j-t} \left[{}_t p_{Nj}^i {}_t y_{Nij} (1-\delta) \right] - b_{it-1} (1+r) \quad (11)$$

In expression (11), ${}_t y_{ij}$ and ${}_t K_{ij}$ are respectively the planned production and capital stock of traded goods in period j by agent i ; ${}_t p_{Nj}^i$ is the expected price of non-tradables in that period, r is the rate of interest, and b_{it-1} is the stock of financial debt outstanding at the end of $t-1$. This equation incorporates the (planned) supply response of the agent to the expected movements in the price of N .

In the perfect foresight case, the individual anticipates the future market-clearing values of the price of non-traded goods. This price, in turn, is generated by the equilibrium between the supply (which depends on the price), and the demand for good N , which is a function of aggregate wealth. Given perfect foresight, moreover, there is no discrepancy between perceived and actual wealth.

It can be seen that the expression for individual wealth (11) can be decomposed in two terms: one which measures the contribution of traded goods, and the other that of non-tradables. A similar decomposition holds in the aggregate. Therefore, aggregate wealth can be written as:

$$W_t = W_{Tt} + W_{Nt} \quad (12)$$

with

$$W_{T_t} = \sum_{j=t}^{\infty} \beta^{j-t} \left[y_j - (K_j - K_{j-1}) - \frac{\phi}{2} \frac{(K_j - K_{j-1})^2}{K_{j-1}} \right] - b_{t-1}(1+r) \quad (13)$$

and

$$W_{N_t} = \sum_{j=t}^{\infty} \beta^{j-t} [p_{N_j} y_{N_j} (1 - \delta)] \quad (14)$$

Along a perfect foresight path, the value of the demand for non-traded goods is a proportion of actual wealth, which remains constant over time. Thus, if agents consume on the basis of the perception that wealth in period t is W_t , it would be the case that:

$$W_{N_t} = W_t h (1 - \delta) \quad (15)$$

Therefore, for a given path of the supply of traded goods, aggregate wealth would be expressed by:

$$W_t = W_{T_t} (1 - h(1 - \delta))^{-1} \quad (16)$$

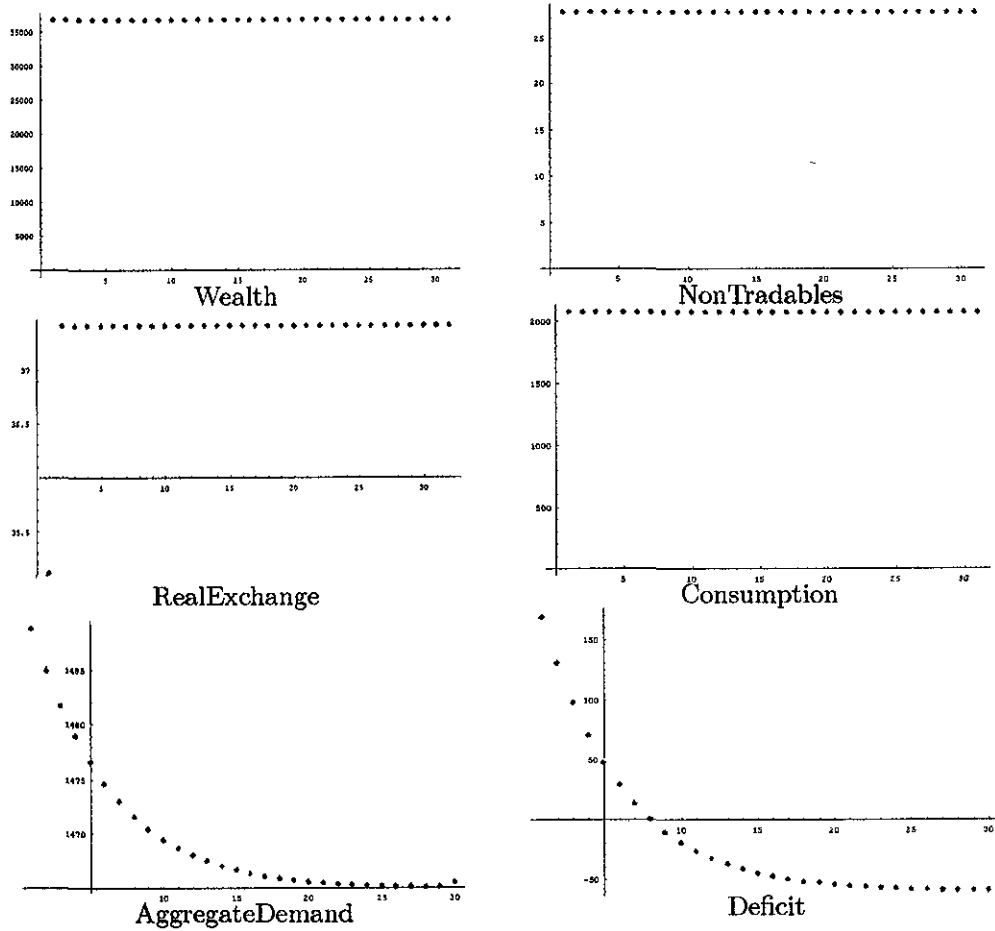
Equilibrium in the market for good N implies a price:

$$p_{N_t}^* = \left[\delta \frac{r}{1+r} (1 - h(1 - \delta))^{-1} W_{T_t} \right]^{1-\delta} \quad (17)$$

No restriction is imposed on the flexibility of the price of N in any direction. **Figure 2** show the perfect foresight response of the main variables of the model after

a disturbance that increases the long-run output of traded goods by around 15 %.

Figure 2



The paths of investment and production in sector T are the same as in the one-good case (this is due to the assumption that T does not use non-traded inputs). Here, too, wealth (actual and perceived) remains constant after the shock. The demand for non-traded goods jumps immediately to a new constant value and, given that the output of good N also adjusts at once, the price of non-tradables goes to the new stationary level: there is a one-shot real appreciation. Once again, aggregate demand rises suddenly, and gradually decreases as the capital stock in T approaches the steady-state. Trade deficits during the first periods are matched by surpluses which tend to converge to a stationary value.

2.3. Learning dynamics

Although the perfect foresight evolution of the hypothetical economy is qualitatively simple, the numerical results depend on the values of various parameters, which can hardly be assumed to be public information. From the point of view of an individual trying to anticipate the future behavior of output for the purpose of drawing his own plans, finding out the path generated by the fundamentals of this simple model of the economy would imply knowing somehow the magnitude of the aggregate shock, the strength of the externality effect, the adjustment costs of investment for the average firm and the elasticity of the supply of non-tradables⁸. These parameters are likely to be difficult to specify in any case, and the more so if the economy is going through a structural change, in which the coefficients of production functions can be supposed to be shifting. Moreover, in order to implement the perfect foresight solution, the information about the parameters should be of common knowledge, so that each agent need not speculate about the responses of others given the fundamentals. This seems indeed a strong hypothesis.

The scenario we are interested in analyzing is one where agents realize that there has been a productivity shock (both for their own specific activity and for the economy as a whole), and they are able to determine in broad terms how the system will react, but where they must learn about the quantitative performance of the economy through some type of heuristic procedure. The choice of such learning scheme is, of course, open-ended. This fact, however, seems to strengthen the argument for assuming that learning is based on heuristic rules (rather than derived from a strictly defined underlying model), since individuals may realize that the observed outcomes depend on the specific way in which average opinions are established, and that this further complicates their own inference problem.

For the purposes of this exercise, we have used a relatively simple learning algorithm to model the way in which individuals update their expectations of the aggregate output of traded goods. This learning scheme seems reasonably well adapted to the case we are studying. As we showed in section 2.1 the perfect foresight model generates a solution with a gradual relaxation of the output of good T towards the steady-state. This solution is approximated by equation (7). We assume that agents form their expectations on the basis of this equation which implies that they can identify the general form of the equilibrium output path: even if they do not try to consider the way in which fundamentals and expectations interact to generate output, they employ a reduced form approach which is appropriate for the problem they are facing. Agents use the observations of actual aggregate output to revise their estimates of the parameters y^* and λ . In period 0, when they receive information about the occurrence of the productivity shock, individuals make a conjecture about the values

⁸In this respect, the agents of the model have both incomplete information in the usual sense (as they cannot ascertain the value of the aggregate shock) and imperfect knowledge of the structural model. However, even if the size of the shock was revealed as public information, the inference problem would remain, since individuals would be unable to specify precisely the aggregate path that would result from the given shock.

of these parameters. Those conjectures are made by processing data that the agents find relevant (drawing from the history of the economy itself, similar episodes of other economies or influential opinions) through some analytical apparatus. In this exercise, we take these initial conjectures as given.

The learning scheme that we postulate is of the stochastic approximation type⁹ (cf. Sargent (1993), Evans and Honkapohja (1993, 1995)). According to this scheme, the coefficients of a function relating a variable y with a vector of variables x , is to be determined by:

$$\begin{aligned}\beta_t &= \beta_{t-1} + \gamma_t R_t^{-1} x_t (y_t - x_t' \beta_{t-1}) \\ R_t &= R_{t-1} + \gamma_t (x_t x_t' - R_{t-1})\end{aligned}\tag{18}$$

The coefficient γ_t can be a function of time. When γ_t diminishes with t (a decreasing gain algorithm), this means that successive forecast errors have less weight in determining the parameters. In a recursive regression, $\gamma_t = 1/t$, and R_t is a standard variance-covariance matrix based on sample values of x_t . The constant gain algorithms result by setting the parameter γ_t at a given, fixed value. Such algorithms will find use when it is believed that the process generating the variable y is subject to frequent changes (cf. Evans and Honkapohja (1995)). In our case, the agent must learn about the parameters y^* and λ of equation (7). The learning procedure starts with initial values for these parameters, and for the elements of the matrix R :

$$R = \begin{pmatrix} 1 & e \\ e & d \end{pmatrix}\tag{19}$$

Then, the matrix is updated with the arrival of new observations by:

$$e_{t-1} = e_{t-2}(1 - \gamma_t) + y_{t-1}\gamma_t\tag{20}$$

$$d_{t-1} = d_{t-2}(1 - \gamma_t) + y_{t-1}^2\gamma_t\tag{21}$$

And the estimated parameters of (7) follow the iterative expressions:

$$y_t^* = y_{t-1}^* + \gamma_t \left(\frac{d_{t-1} - y_{t-1}e_{t-1}}{d_{t-1} - e_{t-1}^2} \right) [y_t - \lambda_{t-1}y_{t-1} - y_{t-1}^*]\tag{22}$$

$$\lambda_t = \lambda_{t-1} + \gamma_t \left(\frac{-e_{t-1} + y_{t-1}}{d_{t-1} - e_{t-1}^2} \right) [y_t - \lambda_{t-1}y_{t-1} - y_{t-1}^*]\tag{23}$$

⁹This recursive method for updating the parameters of a function that determines a variable assumes that the variable is stochastic. Using the procedure in our case implies postulating that the agents view the environment as stochastic, when in fact it is fundamentally deterministic. This assumption has been used in the literature in related contexts. See, for example, Marcet and Sargent (1989).

Equations (22) and (23) indicate that the agent would use the forecast errors that he makes over time in order to recompute his estimates of the coefficients on which he bases the expectations of future aggregate output of good T . In particular, if the individual has underestimated total production this period, he will revise upwards his estimate of long-run aggregate output. The intuition seems clear: the agent interprets the event of a higher-than-expected output as an indication that production will converge to a higher value than he had previously anticipated.

The individual must also form expectations about the sequence of non-traded-goods prices. Now, the path of the real exchange rate depends on the evolution of aggregate expectations. We assume here that agents do not engage directly in drawing expectations of the expectations of others. However, the learning scheme that we postulate implies that individuals try to make consistent expectations of the variables of interest; that is, that they recognize, at least qualitatively, the links that the structure of the model imposes on the magnitudes they have to forecast. Specifically, agents take a fundamental view of the long-run level of the price of N , and possibly presume that the actual price will move gradually to that fundamental value. The forecasts of the price of N that the individual uses in order to plan output and consumption would then be generated by:

$${}_tP_{N_j}^i = \mu P_{N_j-1} + (1 - \mu) {}_tP_N^{*i} \quad (24)$$

Here, ${}_tP_{N_j}^i$ denotes the expectation formulated at time t by individual i of the price of non-traded goods at time j , and ${}_tP_N^{*i}$ is the fundamental price as perceived at t by agent i (before observing the aggregate variables for period t). The adjustment parameter μ (which may be zero) measures the degree of persistence that the agent attributes to prices in their movement towards the perceived fundamental price. This price, we assume, is estimated by the agent realizing that the demand for non-traded goods depends on the current and future supply of tradables. Given the parameters y_t^* and λ_t estimated with information up to period $t - 1$, the agent can project the sequence of the aggregate output of traded goods. He can also use the production function to find the aggregate capital stock that would be consistent with the steady-state production of T that results from the sequence (7)¹⁰. We postulate that the individual generates the path for aggregate investment using the same adjustment coefficient λ as in (7). The individual is now able, after observing the aggregate foreign debt at the end of period $t - 1$, to make an estimate the contribution of traded goods to the economy-wide wealth (using an expression analogous to (13)):

¹⁰At this point, we endow the agent of knowledge of the production function. We will assume below that he also has information about the adjustment costs of the capital stock. It seems preferable to make these (somewhat inconsistent) assumptions that to introduce more learning schemes for variables that are not crucial for the analysis. The possibility of misperceptions in the agent's estimate of aggregate wealth appears directly if he has no way of forecasting accurately the actual path for the output of traded goods.

$${}_tW_{T_t}^i = \sum_{j=t}^{\infty} \beta^{j-t} \left[{}_ty_j^i - ({}_tK_j^i - {}_tK_{j-1}^i) - \frac{\phi({}_tK_j^i - {}_tK_{j-1}^i)^2}{2{}_tK_{j-1}^i} \right] - (1+r)b_{t-1} \quad (25)$$

where ${}_tW_{T_t}^i$ is the aggregate wealth derived from traded goods, as perceived by agent i at time t ; ${}_ty_j^i$ and ${}_tK_j^i$ are, respectively, the aggregate output of T and the aggregate capital stock at time j that agent i projects at time t ; b_{t-1} is the actual foreign debt at the end of $t-1$. From this, the agent forms an estimate of what would be total wealth if aggregate consumption plans were to be based on the estimate of sustainable wealth that the agent himself makes. By analogy with (15), this would be:

$${}_tW_t^i = {}_tW_{T_t}^i (1 - h(1 - \delta))^{-1} \quad (26)$$

This expression implicitly defines the sustainable demand for non traded goods in period t as perceived by agent i . If, furthermore, the agent assumes that the sustainable price of N is generated by spending decisions resulting from wealth estimates (26) and the profit maximizing output of N at the sustainable price, we obtain,

$${}_tp_{N_t}^{*i} = \left[h\delta \frac{r}{1+r} (1 - h(1 - \delta))^{-1} {}_tW_{T_t}^i \right]^{1-\delta} \quad (27)$$

Once the variable is determined by equation (27), the individual has all the inputs to form his expectations and solve his decision problem. It may be noted again here that, in the two-goods setting, the perceived wealth of a given agent depends on his expectations about the future output of tradables, through its effect on the demand for the non-traded goods that he produces. Therefore, perceptions about the aggregate performance will influence market behavior, even in the absence of externalities.

To summarize: Given the way in which the agent makes his forecasts, the exercise proceeds thus (see flow diagram 1). In period $t-1$, the individual observes aggregate output, and computes the values of y^* and λ . These values generate a sequence of output forecasts (for $t, t+1, t+2, \dots$) that will be used to draw a plan at t . In this period, the agent employs the forecasts to solve a production-accumulation program (through equation (4)). The solution to that program (together with the size of the financial liabilities accumulated up to period t) generates the value of the traded-goods component of the perceived wealth of the individual. As explained above, the agent, using the forecasts he has made for the future total output of traded goods, computes the aggregate sustainable wealth according to equation (26). He then can estimate the sustainable price of non-tradables (through (27)), and build an expected sequence of the price of N (using (24)). With these estimates, he can formulate a plan for the output of non-tradables. This plan determines the non-traded goods component of perceived wealth. Using equations 11 through 13, the agent now estimates his total wealth, from which the consumption plan is derived.

The individual spends (in consumption and investment), produces non-traded goods and borrows or lends in period t according to the plans he has formulated¹¹. Output of traded goods in t is pre-determined by the capital stock in place at the end of $t-1$, but it is observed only in t . The debt of the agent increases by the difference between planned (equal to realized) spending in period t and income (net of interests on the pre-existing debt). Given the spending and production plans implemented at period t , the market-clearing price of non-tradables is determined. After observing output, the agent now updates again the estimates of α and η , which are fed into the decisions taken at $t+1$. The sequence of actions which are executed in the successive iterations generates the solutions of the model.

2.4. Simulation results¹²

As mentioned before, the experiment consists in shocking the system at $t = 1$ (the shock being perceived by the agents when the plans for that period are still to be drawn) with a permanent productivity increase in the traded goods sector, such that the steady state output of that sector rises by around 15%. Given the parameters used in the exercise (particularly the adjustment cost coefficient), under perfect foresight, the production of tradables would show moderate rates of growth during the transition between steady states, starting with an increment of around 2.6% in period $t = 1$; nearly 85% of the total adjustment in output would take place in the first 10 periods. Total wealth increases by about 9%, inducing a one-shot real appreciation of about 5% and a 3.5% rise in the production of non-traded goods. The foreign debt that is accumulated during the transition grows to a sizeable proportion of GDP (around 45%), which is serviced with the trade surpluses which are generated from period 9 on¹³.

For the appropriate choice of the initial parameters, the model that incorporates learning can replicate quite exactly the perfect foresight path¹⁴. That is, if individuals use the learning algorithm specified in the previous section, and they start with a correct perception of the long-run level of output and the speed of convergence, and they assume that the price of non traded goods will go immediately to its sustainable value, the observed outcomes will validate their expectations (cf. **Figure 3**). Therefore, the successive updates of the parameters will confirm the initial beliefs. The decisions that the agents make under those expectations are almost indistinguish-

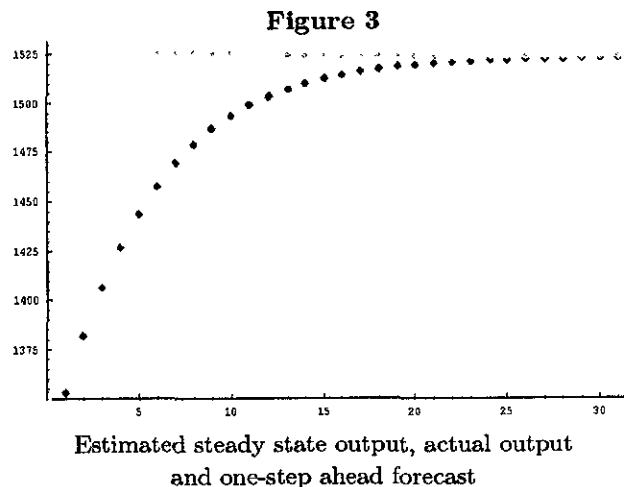
¹¹It may be worth noting here that only the immediate decisions of a given intertemporal plan are actually implemented. In every period, the agent revises his forecasts and solves a new set of optimization problems.

¹²The values for the basic parameters used in the simulation are the following: $\alpha = 0.65$, $\eta = 0.25$, $\phi = 0.5$, $r = 0.06$, $z = 2.45$, $h = 0.5$ and $\mu = 0.2$.

¹³Needless to say, the simulation experiments are not designed to represent the behavior of any actual economy, but to provide a framework to analyze the qualitative effects of the learning process. The references to the numerical outcomes of the simulations are simply meant to illustrate the features of the results that the model generates.

¹⁴Here, as in all the experiments that follow, the matrix R_0 is determined with $e = 0.9y_0$ and $d = 1.1y_0^2$, where y_0 is the pre-shock level of traded-goods output.

able with those that are obtained under the assumption that the agents are actually informed about the future evolution of the aggregate variables.



The exercise indicates that the learning scheme is indeed a reasonable one, in the sense that, given the postulated forms of the functions agents use to make forecasts, there exists one set of starting guesses which remain practically constant under learning and which will result in expectations closely tracking the observed results.

Although the model can be used in principle to simulate different scenarios, we will concentrate first on the case where the initial conjectures of the representative agent are such that his estimate of the steady-state aggregate output of traded goods is higher than the perfect foresight solution, and he anticipates a slower convergence to the steady-state than would apply on the perfect foresight path¹⁵. That is, the agent is over-optimistic regarding the long-run value of total production, but he has an exaggerated perception of the time that the transition will take (and, therefore, he does not form too-high expectations about the short-run increase in the production of traded goods).

The following figures summarize the simulated behavior of the system, for the case where the representative agent starts with the belief that the output of traded goods will eventually rise by 10% above the perfect foresight steady state after the shock, and where the initially expected speed of convergence is given by a parameter $\lambda = .95$, instead of the value (near 0.83) that approximates the behavior along the perfect foresight path. These parameters are updated with a constant gain algorithm, with $\gamma = .5$. The expectations for the price of non-traded goods are based on equation (24) with a parameter $\mu = .2$, indicating that the agent puts much weight on his estimate of the fundamental price.

¹⁵For the sake of parsimony, we only report here the results of some of the simulations that were calculated. A more thorough description, together with a more detailed exposition of the procedures used to run the model will be presented in a separate paper, to be written jointly with P. Azcue.

Figure 4

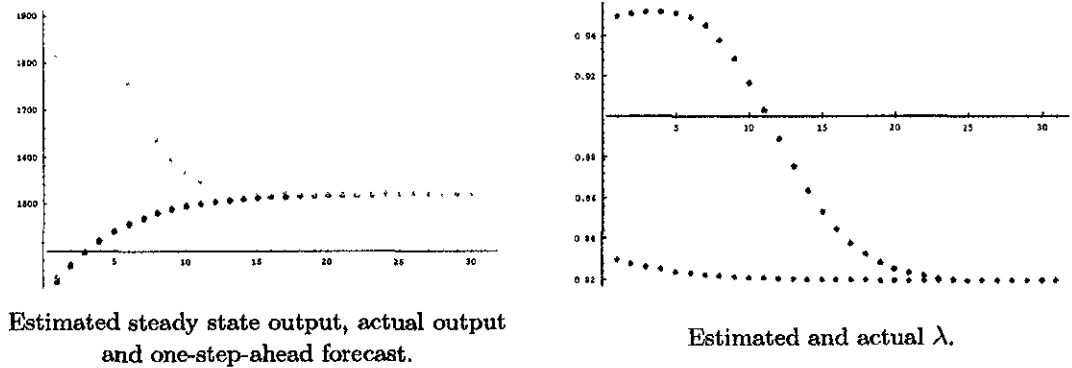


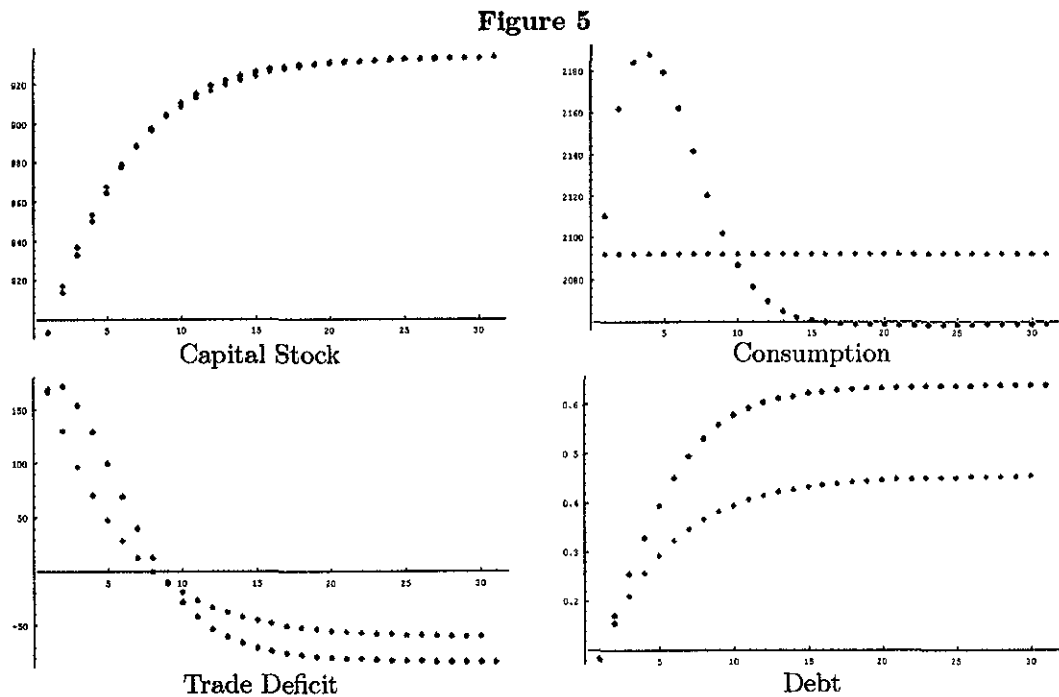
Figure 4 (left panel) shows the evolution of three variables: the expected steady state level of output of tradables generated by the learning algorithm, the actual level of output, and the one-step-ahead forecast of this variable that results from using equation (7) with the parameters updated with the information available at the end of the previous period. Several features may be noted:

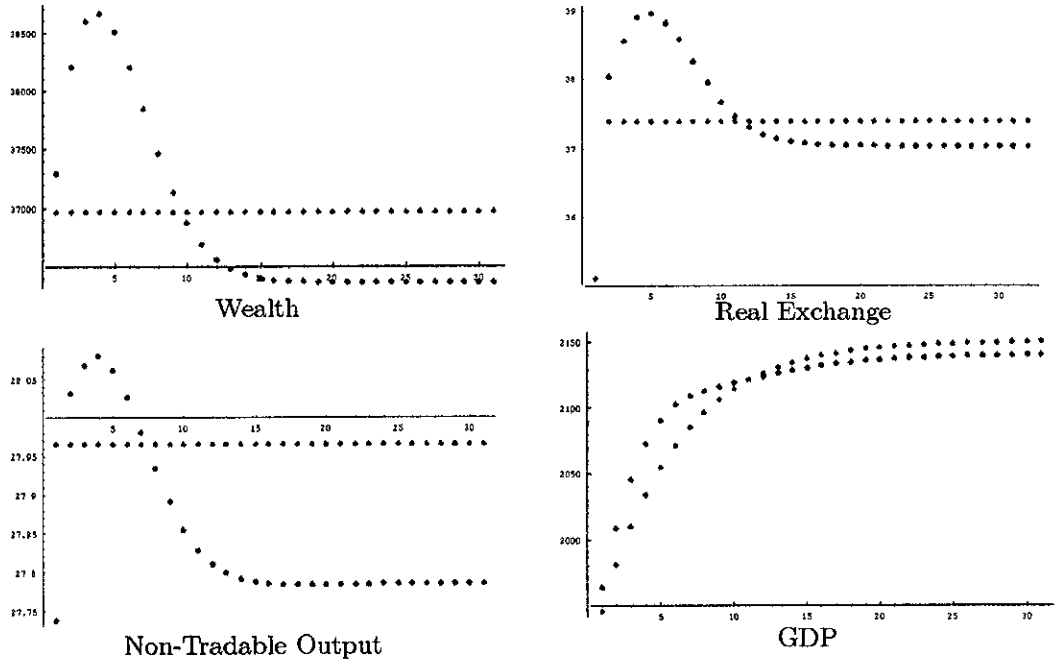
1. The estimates for the steady state output follow a non-monotonic path: they first increase for some periods (thus raising the discrepancy between the expected and the actual value), and later gradually relax to a value that is practically equal to the level to which actual output converges. In other words, the learning scheme eventually makes agents aware of the asymptotic value of the total production of tradables, but errors are not corrected at once; rather, after observing the aggregate realizations, initial mistakes may be amplified for a while.
2. The reason for this behavior is clearly that, given the initial beliefs of the agent, the aggregate production of tradables outperforms for a certain number of periods relative to expectations, leading the individual (mistakenly) to revise upward his perception of long-run output.
3. Although the agent may be wide of the mark in his long-run estimates, the short-run forecasts that the model delivers are reasonably accurate. The absolute value of the errors in the one-step-ahead forecasts of the level of (traded) output never exceed .9%, and fall below .2% after the seventh period. In terms of growth rates, the agent makes a somewhat large error in the first period (for which he forecasts a 1.4% increase, when the actual value is 2.3%), but by period 3, the discrepancy falls to less than .2 percentage points. This implies that learning about the actual value of steady state output will be slow (since it is

driven by the period-by-period expectational errors) . The results suggest that mistakes about long-run trends may show persistence, given that they are not necessarily contradictory with good forecasting performance in the short-run.

Figure 4 (right panel) also indicates the movements of the parameter λ under learning, compared with the one that would be calculated along the perfect foresight path. The figure shows that the estimated parameter eventually converges, but not too rapidly.

The graph for the capital stock in the traded goods sector with perfect foresight and under learning can be seen in **Figure 5**. It can be observed that there is indeed convergence to a steady-state, and that the values of the capital stock and aggregate output after the transition tend to those that would apply under perfect foresight.





However, aggregate demand fluctuates. In particular, aggregate consumption (measured in terms of traded goods) rises above the perfect foresight value and keeps increasing for some periods (in this particular case, reaching a maximum in period 4, 5% over the perfect foresight benchmark). After a while, consumption declines, and tends to a value that lies below that of the perfect foresight path. The reason for this long-run effect of the cumulative forecast errors is, of course, that the initial over-expansion of aggregate demand causes the foreign debt to rise excessively: the transitional misperception of wealth induces a permanent rise in debt services, so that consumption must adjust in order to generate the corresponding trade surplus and satisfy the intertemporal budget constraint.

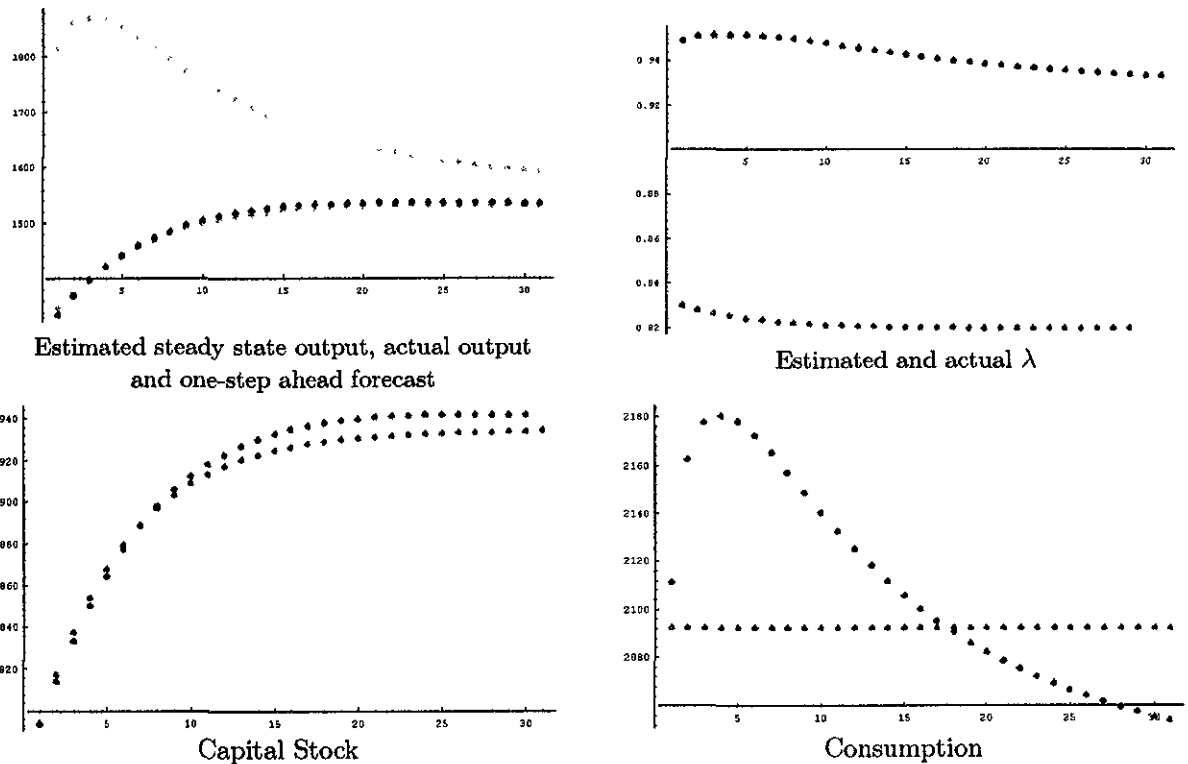
The simulations also indicate that the wealth misperceptions are not corrected immediately; rather, they get amplified for some periods. That behavior is also reflected in the path of relative prices and in the output of non-traded goods. The initial real appreciation exceeds the perfect foresight level, and the price of non-tradables keep rising for some periods (until a peak that corresponds in this exercise to a 5% misalignment¹⁶). The gradual process of depreciation that follows makes the price of non-tradables go lower than the value that would have resulted with correct

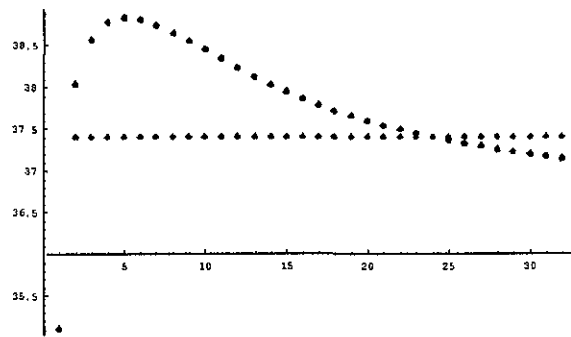
¹⁶In this exercise, the term is well defined since, by construction, there is a perfect foresight value on which to base the comparison.

expectations throughout the experiment, but this price remains noticeably higher than at the pre-shock steady state.

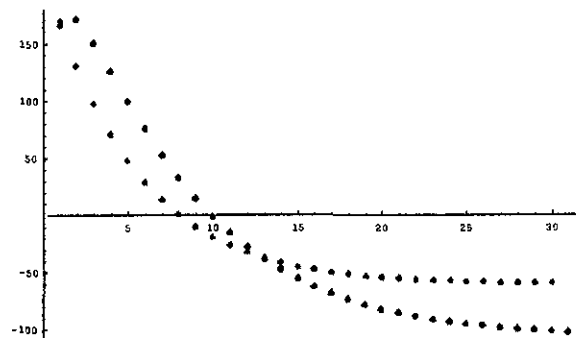
Figure 6 shows similar simulations using a decreasing gain algorithm where the successive weight of the forecast errors for the purpose of revising the parameters gets smaller as the time interval after the shock grows. We see that the convergence of the parameters is slower than in the previous case. Although the learning scheme makes the estimated parameters tend eventually towards their steady state values, after 30 iterations there are still non-negligible discrepancies, particularly in the case of the coefficient λ .

Figure 6

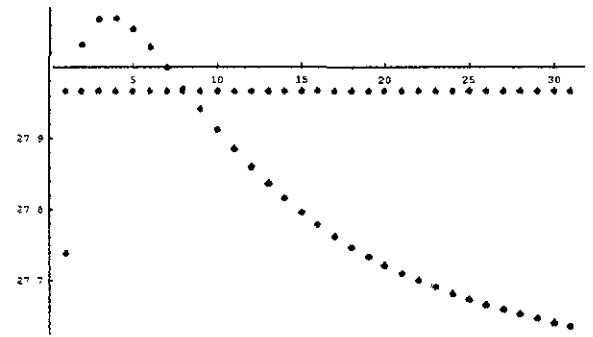




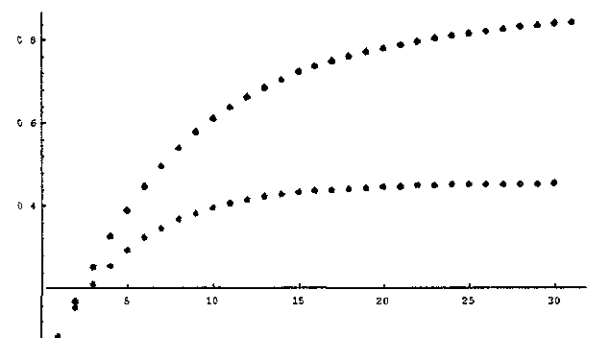
Real Exchange



Trade Deficit



Non-Tradable Output



Debt

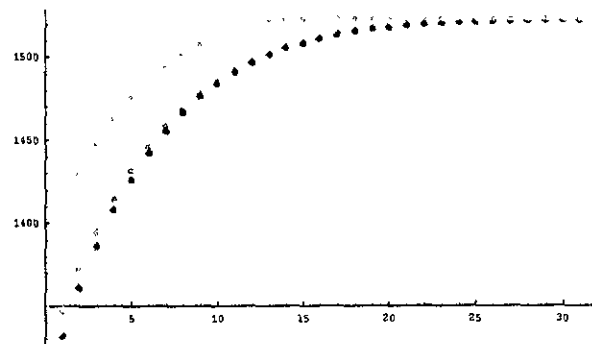
Thus, the model can generate smooth cycles in spending, relative prices and the production of non-traded goods¹⁷. Such cycles arise for some configurations of the initial beliefs of agents. The dependence on initial conjectures is to be expected in a consistent account of episodes of spending overshooting: if agents could easily find out that the system moves along a path that implies an intertemporal coordination failure, they would revise their decisions. The misperceptions should be hard to identify as such in order to remain uncorrected for some time.

It can be verified that, if agents initially underpredict the steady state level of traded goods output and they overpredict the speed of convergence to the stationary value, the system does not undergo fluctuations. This is shown in **Figure 7**¹⁸.

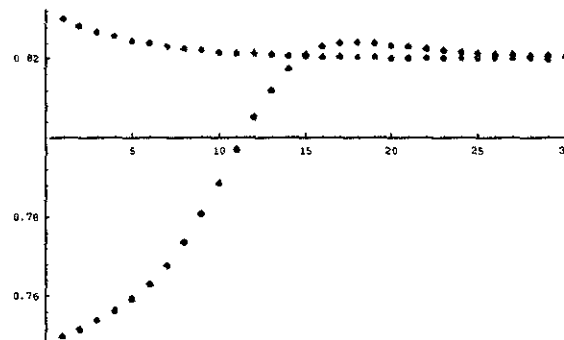
¹⁷The smoothness of the fluctuations does not describe well the recessions that follow balance of payment crises. There are two related reasons for that feature of the model. As successive observations show agents that their wealth perceptions were too high, the revision of beliefs takes place gradually, and access to foreign financing stays unchanged during all the process, with an elastic supply of funds at the fixed international interest rate. Clearly, a good representation of the downward phase would require specific consideration of the behavior of financial markets. In this regard, the model is more apt for the analysis of the upswings than of the entire cycle.

¹⁸The simulations correspond to the case where the starting estimate of the steady state output of traded goods is 10% below the perfect foresight level, and the initial guess for the speed of convergence corresponds to $\lambda = 0.75$. The learning scheme follows a constant gain algorithms.

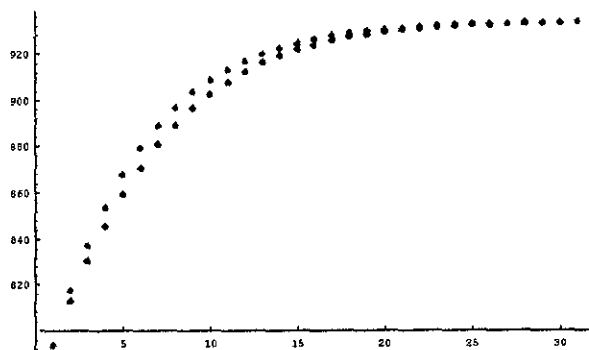
Figure 7



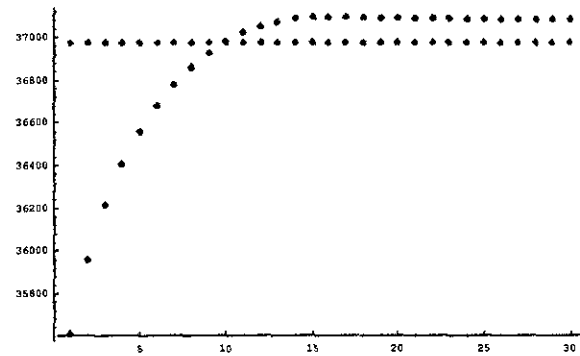
Estimated steady state output, actual output
and one-step ahead forecast



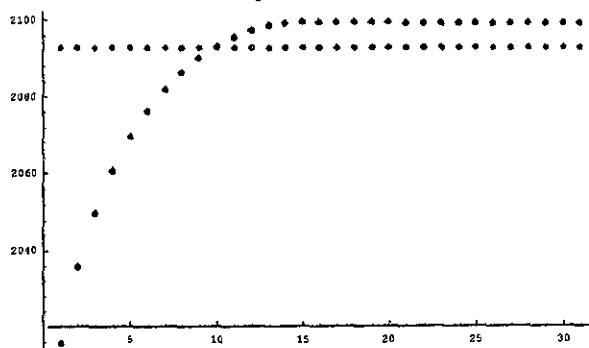
Estimated and actual λ



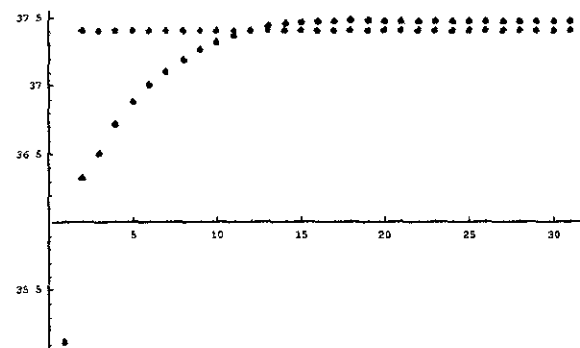
Capital Stock



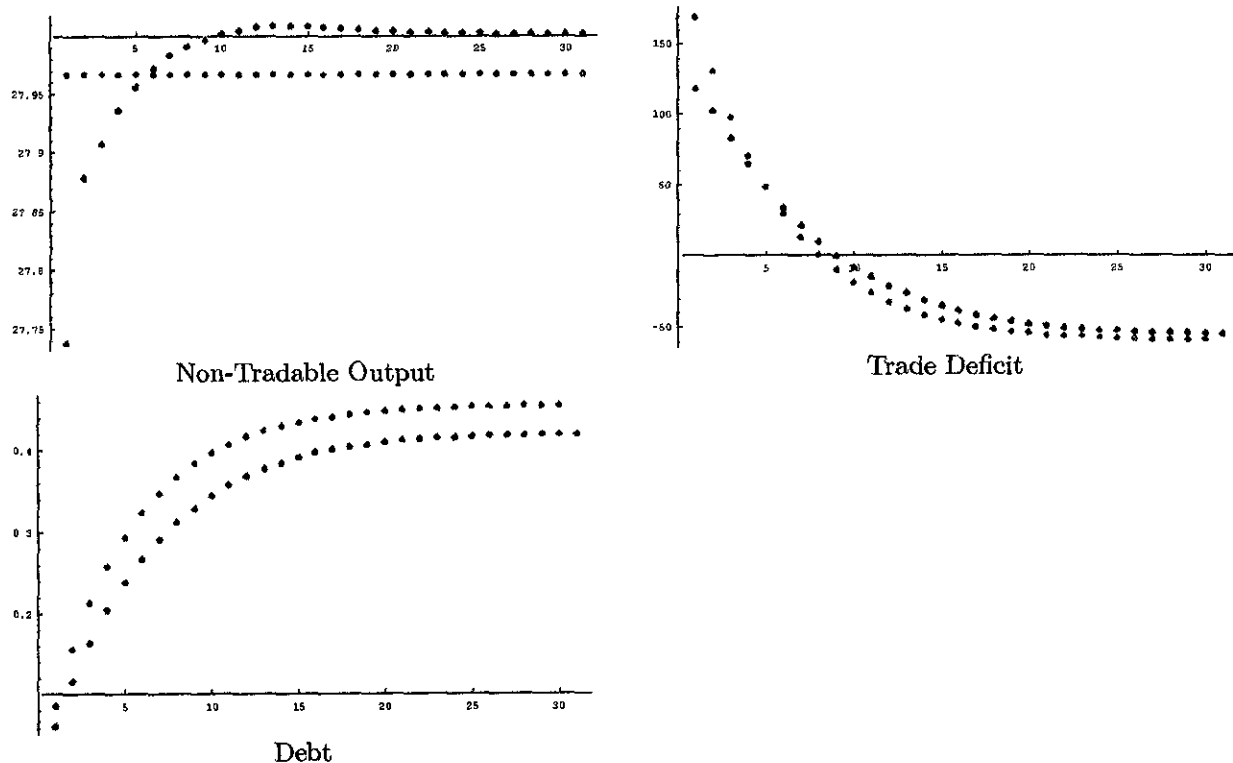
Wealth



Consumption



Real Exchange



Here, the agents start with too low wealth perceptions, which are adjusted gradually, always in an upward direction. Consumption, and the production and price of non-traded goods all start from values below the perfect foresight benchmark, and converge monotonically to a level that is higher than this benchmark. The last result is due to the fact that during the transition aggregate spending is such that agents accumulate a smaller foreign debt than under perfect foresight: when the error (this time one of insufficient optimism) is realized, the agents have lower liabilities and therefore a higher capacity to spend.

The model is obviously very simple, and its parameters have not been calibrated so as to approximate any actual economy. However, the results delivered by the simulation seem suggestive of some types of behavior which may be observed in practice. A positive shock will not necessarily induce an overshooting in aggregate spending; reciprocally, there is no a priori reason to deny the possibility. In certain circumstances, agents cannot easily identify the long-run behavior of the system implicit in its fundamentals. But learning about the longer-run performance of the economy is (almost by definition) a process that takes time. The signals which are received period-by-period need not be given a correct (as seen ex-post) interpretation, even

though agents use reasonable learning schemes. Transitional cycles can be generated as a consequence.

3. Misperceptions as temporary equilibria

The previous section analyzed a model where agents learn from experience, after a shock which disturbs the expectations they previously held. Individuals have to generate forecasts of the future path of the economy in order to make their current decisions, but they must rely on scarce direct information (i.e. a small sample of actual observations) about how the economy will behave on the new growth path. They have to rebuild a working model with which to form expectations, based on whatever prior knowledge they possessed about the likely effects on the shocks, and their interpretation of the observed outcomes. In those circumstances, it seems natural to assume that the individuals engage in some sort of heuristic learning. The resulting process of revision of beliefs may be such that errors of judgment are not automatically self-correcting.

In that setting, individuals used the discrepancy between expected and realized outcomes to adjust their forecasting model; their learning procedure was not conditional on beliefs about how other agents form expectations. Here, we consider a case where individuals try to analyze the behavior of others as expressed in the markets. The main question that we ask is whether a state with misperceptions can be sustained as a temporary equilibrium.

The general motivation for the exercise is the following. Individuals have good reasons to condition their expectations of future income on their beliefs about the behavior of the economy as a whole. Growth in aggregate output increases the sustainable demand for individual goods and services, and it widens productive opportunities. In addition, the behavior of other agents, as reflected in their decisions to produce and to spend, may inform a given individual about how to react to the signals that he observes, but which by themselves may have a diffuse meaning. Now, in practice, increases in aggregate demand are often subject to alternative interpretations, either as signs of fundamental improvements in economic conditions, or as indicators that spending has become excessively high. If individuals strongly believe that the first interpretation is valid whatever the circumstances, rising aggregate expenditures would then lead them to form more optimistic expectations of the economy's performance. To the extent that wealth perceptions react positively to those expectations, individuals would then plan a higher consumption. Then, it may conceivably happen that exaggerated expectations produce market information which confirm the initial mistaken beliefs. This is the possibility that we want to explore.

As an example, we consider the case of two individuals (or two sets of agents which are alike within each group and recognize the intra-group homogeneity). The future income of agent i depends on two variables: one that he observes directly in the current period, and another which is observed by agent j (but not by i). That is:

$$y_i = g(z_i, z_j) \quad (28)$$

The dependence of y_i on z_j can be rationalized in two ways: as a direct effect of a variable pertaining to agent j on the future opportunities of agent i (say, through an externality), or as an informational effect, if a signal received by j carries information that is relevant for i . We will first analyze the limit case where the function is symmetric in its arguments: $g(a, b) = g(b, a)$. The future income of agent i then depends as strongly on the signal that j receives than on his own signal; that means that the externality has quite a large influence on individual incomes.

For concreteness, suppose that the economy has received good news, which raise the individual estimates of future income. These news are incorporated in the signals that each agent receives. However, these individual signals do not fully specify the effect on future income. Each agent can receive, say, either a weak (z_w) or a strong signal (z_s). Given his own signal and the estimate that he makes of the signal of the other agent, each individual decides a level of consumption (this is the only choice variable). Aggregate consumption is publicly observed. The agents know the form of the functions relating consumption to expected income. We will consider the possible states where the beliefs of both agents are validated by the observation of aggregate consumption.

Suppose that both agents take as a certainty that the other one has made a correct decision, and assume that both have received a signal z_w . Then, clearly, the consumption level determined by the expectation $g(z_w, z_w)$ on the part of each individual is an equilibrium as defined before. Agent 1, for example, when observing that agent 2 expects his income to be $g(z_w, z_w)$, infers that the signal received by 2 has been z_w , since he imputes to agent 2 the knowledge that 1 has had a z_w signal. Therefore, the correct level of consumption is self-validating.

However, this is not the only possible equilibrium. Suppose now that both agents have received a weak signal but, for some reason, they both come to expect that the other agent has received a strong signal. In that case, they would both consume now on the basis of the expectation that future income would be determined by a (w, s) configuration, when in fact it will be given by (w, w) . Aggregate consumption would then be unsustainably large. But, if agents have a firm belief that market outcomes reflect accurate perceptions of future income, they would interpret the (too large) increase in consumption as an indication that their initial assumption was justified: agent 1, say, observes agent 2 consuming as if the future state of the economy would be given by (s, w) and, if 1 judges that 2 has realized that the signal for 1 has been w , he feels confident that the signal for 2 has been s , as originally conjectured. Somewhat ironically, it is the confidence that agents have on the predictive ability of others (and of the market as whole) that makes them persist in forming wrong expectations. In this temporary equilibrium, both individuals have an overoptimistic perception of future income, since they wrongly anticipate that they will be benefitted by a strong externality from the other, who is expecting the same. The confusion appears because each agent attributes the high consumption of the other to a strong observed signal

and not, as it is the case, to inconsistent expectations.

The example generalizes without too many changes to the case where there is a large number of different agents, under the assumption that the future income of agent i depends on the average of the signals received by the individuals in the economy, i.e. that the income of a given individual moves together with aggregate income. Suppose, as before, that agents can individually receive either a strong or a weak signal, and let f the fraction of agents observing signal z_s . It is clear that aggregate consumption will depend on the perceptions of the agents about the fraction f . If individuals make their estimates of f equal to the average estimate as indicated by aggregate consumption (because they believe that the level of demand reveals the magnitude that is sustainable, given the fundamentals, in this example, the coefficient f), the system would be in an indifferent equilibrium, where any (collectively) perceived value of f would be self-validating, whether it is or not the true value. Once again, the confusion arises because individuals deny the possibility that aggregate consumption may differ from its sustainable value: they identify actual future output with the average expectation of future output.

Clearly, the assumption that only the average signal matters for the determination of an individual's future income is rather strong. If the specification is modified so that an agent's income depends both on the own signal and the average one, the result that belief in the accuracy of market predictions sustains aggregate errors is no longer literally valid: the introduction of the individual signal as an argument of future income breaks the symmetry of the formulation and, given the knowledge they are supposed to have in the exercise, the inference problem has no longer multiple solutions, if all agents make the same conjectures.

We then specify now future output of individual i by:

$$y_i = g(z_i, \bar{z}) \quad (29)$$

As before, a proportion f of the agents observes signal z_s . If f' is the (unanimous) perception that individuals have of f , the average expected individual income is:

$$\bar{y}^e = fg(z_s, f'z_s + (1 - f')z_w) + (1 - f)g(z_w, f'z_s + (1 - f')z_w) \quad (30)$$

It is clear that, if $f' > f$, the average expected income (and, therefore, aggregate consumption) is lower than the value that would sustain the perception $f = f'$. In other words, the average of individual income expectations (which is revealed by consumption) is lower than the average of the individual expectations of average income (which each agent uses, together with his own signal, to forecast his income and decide his own consumption). That is, given their belief that market expectations are accurate and that the fraction of agents with signal z_s is f' , individuals would expect to observe:

$$(\bar{y}^e)^e = f'g(z_s, f'z_s + (1 - f')z_w) + (1 - f')g(z_w, f'z_s + (1 - f')z_w) \quad (31)$$

The only self-sustained unanimous perception is now $f' = f$, the true value, since only then $\bar{y}^e$ and $(\bar{y}^e)^e$ would be equal. In the case where agents initially overestimate f , a relatively low level of consumption would inform them that the average signal is smaller than they expected. It seems worth mentioning that the revision of beliefs would not come about in this case because individuals judge that there has been an overexpansion in aggregate demand, but rather the opposite: in order to sustain their optimism, they would have to observe that the average trader is consuming more than he is. Suppose, however, that the average individual is overoptimistic, but that he believes that the market outcome reflects a too-prudent behavior. In this context, that would mean that the individual bases his consumption on the view that the fraction of agents with signal z_s is $f' > f$, but he also thinks that the average expectation is $f'' < f$. Then, the possibility of self-sustained misperceptions reappears, if:

$$\bar{y}^e = f g(z_s, f' z_s + (1 - f') z_w) + (1 - f) g(z_w, f' z_s + (1 - f') z_w) \approx (\bar{y}^e)^e \quad (32)$$

$$(\bar{y}^e)^e = f' g(z_s, f'' z_s + (1 - f'') z_w) + (1 - f') g(z_w, f'' z_s + (1 - f'') z_w) \quad (33)$$

In this case, the individual interprets that the observed magnitude of consumption reflects strong fundamentals (in fact, stronger than they really are) and, at the same time, he thinks that aggregate spending fails to be as high as those fundamentals would allow because the other agents have not fully incorporated that fact into their own expectations. A similar result may hold when, for example, agents still believe that average expectations are correct, but aggregate consumption does not fully reveal those expectations. This can happen, say, if consumers are (or can be perceived as being) credit rationed. In that situation, the individual who bases his own expectations on the level of aggregate demand can judge that the expansion is limited, not by the available productive opportunities, but by a liquidity constraint. In a similar vein, a misperception can be sustained if the average individual takes what is actually an excessive consumption boom for a risk-averse response to strong productive signals.

Throughout this discussion, we have treated all agents symmetrically. However, asymmetries in the ability of agents to interpret signals, and in the confidence with which they hold expectations seem important in practice. In this regard, expectational biases of influential agents may possibly have at times strong effects through the informational content of their pronouncements or their actions.

4. Concluding remarks

The models with which we have dealt in this paper are of a very schematic nature. We would suggest, however, that they may have some practical relevance for the analysis of open economies which undergo cycles of aggregate demand booms followed

by adjustments. The interpretation of these episodes as symptoms of intertemporal coordination failures seems less strained than either the arguments based on perfectly foreseen fluctuations—where the adjustment appears merely as a phase of optimal plans drawn from the beginning of times—or the arguments which rely on purely random shocks—in which case the cause would have to be searched in some type of natural or at least extra-economic event since, if the rational expectations hypothesis is to be taken at face value, agents would make their forecasts with a model that encompasses as endogenous variables, say, the shocks originating in foreign economies. It seems simpler to state that, faced with an environment which produces hard-to-interpret information, individuals make mistakes which are reflected in the aggregate evolution of the system. It should be stressed that this implies, precisely, that out-of-equilibrium fluctuations are difficult to predict (as opposed to rationalized ex-post), both for the agents populating an economy and for the analysts that observe it. However, such difficulty has concrete implications: it is not the same thing to act (or to reason) as if intertemporal coordination failures were impossible than to recognize the possibility. This holds likewise for private agents, for policy authorities and for analysts: taking into account one's cognitive limitations can be seen as a form of rationality.

The arguments that we present have some corollaries which it may be worthwhile to mention. First, the potential for cyclical behavior of different economies can depend on the predictability of the growth path. Second, while the consequences of the lack of credibility of economic policies have been very much emphasized in the literature, too much confidence may also cause disturbances (this point is also stressed in McKinnon and Pill, *op. cit.*). In our models, spending cycles arise because agents are (perhaps implicitly) over-confident, in the complementary senses that they have exaggerated expectations about the steady state level of income, and that they misinterpret increases in aggregate demand as signs that reinforce their optimism, thus undervaluing the chances of a future downturn. From this perspective, the analytical practice of viewing any movement in aggregate variables as part of a well-coordinated, rational expectations path can well contribute in some cases to strengthen inconsistent beliefs on the part of economic agents.

Third, an effect reminiscent of the Lucas critique may apply for policies of economic reform, although the argument would not be based on the hypothesis that expectations are correct, but rather on the assumption that they are forward-looking, though possibly mistaken. Consider a similar set of policies being applied successively in two different (but roughly comparable) economies. In the first experiment, the effects of the policies in stimulating growth may come as surprise. In the transition, individuals would then underestimate their wealth, and consumption would be too low with reference to a rationally expected path. While the rise in income outruns expectations, the economy would then show high savings rate (and growth can appear to be export led). If the agents of the second economy are aware of the experience of the first one, their behavior can well be different (especially if, as it can be the case, the policies are presented to the public as instruments to reproduce the other

economy's success). Their estimates of sustainable spending (and the chances of debt repayment as evaluated by prospective lenders) may incorporate the expectation of large increases in future income. Savings would then be low. The second economy can grow faster than the first one (since the difficulty in establishing credibility would be less severe), but it may also happen that, while the first economy has shown steady growth, the second goes through a boom-adjustment cycle.

Even though the reform packages look alike, macroeconomic policy-makers in these two hypothetical economies face different problems. In the case of the first one, the main issue may be that of overcoming an initial skepticism on the part of the public, which can delay growth. Policy-makers in the follower economy would have to consider the symmetric problem, that is, that of avoiding states of excessive spending (or, to put it differently, states of insufficient savings and too high current account deficits). This illustrates a more general point. Whether explicitly or implicitly, macroeconomic policies are predicated on assumptions made by the authorities regarding the economy's growth potential, and about the expectations that the private sector is forming at the same time. In economies under more or less rapid change, the choice of the adequate policy mix seems hard to reduce to the application of some mechanical procedure.

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